



RICCARDO PERIN

Biomedical Engineer

CONTACTS

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PROFILE

Master's graduate in Bioengineering with a strong focus on Machine Learning and predictive modeling. I am particularly passionate about its application in healthcare and prevention.

I have a proactive approach to tasks and responsibilities. I am highly motivated and excited to continuously expand my knowledge and contribute in a diverse team environment.

LANGUAGES

Italian (native)
English (IELTS 8.0 - C1 advanced)

TECHNICAL AND SOFT SKILLS

- Languages:** Python, MATLAB, R, HTML/CSS, Flutter
- ML/DS:** scikit-learn, pandas, numpy, PyTorch
- Data:** SQL, feature engineering, data pipelines
- Soft:** adaptability, continuous and independent learning, communication and team-work, problem solving and time management

EDUCATION

MSc in Bioengineering - Biomedical Data Analysis and Modeling, University of Padua, Italy,
2025 - present

BSc in Biomedical Engineering (110/110 - highest distinction)
University of Padua, Italy,
2022 - 2025

RELEVANT UNIVERSITY COURSES

- Machine Learning and Data Science (30L/30)
- Neural Networks and Deep Learning (30L/30)
- Foundations of AI (ongoing)
- Methods for Pharmacometrics(ongoing)
- Biomedical Image Analysis (30L/30)
- Signal processing (30L/30)
- Biomedical Data Analysis (ongoing)
- Computational Biology (29/30)
- Modelling Methodologies for Physiology and Medicine (26/30)

RESEARCH EXPERIENCE

Research Collaborator – DIANA Project @ UNIPD (ongoing)

Collaborating with Professor Giacomo Cappon on the DIANA Project (Digital Twin and AI for adaptive management of pediatric Type 1 Diabetes under Multiple Daily Injections therapy).

- Actively developing a machine learning-based meal detection model from continuous glucose monitoring data, currently in the exploratory/prototyping phase.

Deep Learning Project

Development of a deep learning network for brain tumor classification from MRI imaging

- Creation of three models (CAE, ResNet50 and ViT)
- Comparison of the models and use of GradCam to explain how the model was working
- Best model with an accuracy of 95.25%

Bachelor's Thesis research

Development of a Predictive Model for Neonatal Intensive Care Unit admission based on Random Forest and Clinical Record Data

- Performed end-to-end ML pipeline development: data preprocessing, model selection, hyperparameter optimization and performance evaluation
- Investigated class-imbalance mitigation strategies to improve sensitivity on the positive cases
- Final model achieved 66% of F1-score and AUC=80%

RESEARCH INTERESTS

My core research interests lie at the intersection of Artificial Intelligence, Data-Driven Health and Pharmacometrics Modeling (PK/PD/NLME). I am actively looking for PhD opportunities in these fields.